LESSON PLAN

UPDATED 12 June 2019

Name: **Cohort:**

Lesson Plan Day 1

Lesson Title: How Do Gear Ratios Influence, Torque, Speed, and Mechanical Advantage?	**Date:**
Grade Level: 11U ***(12C Modifications)***	**Subject/Strand:** Physics Grade 11U Physics: Strand C (Forces) & Strand D (Energy and Society) ***Not going to include the grade 12 into here just yet since I think we only have to do the plan for one of the grades***
Topic: Gears, Torque, Mechanical Advantage	**Length of Period: 75 minutes**

Lesson Plan Description -- What do I want the learners to know and/or be able to do? What are the big ideas/enduring understandings for the lesson/unit? Indicate using 1-2 sentences.

Students will design, construct, and test an initial version of a gear-driven elevator system capable of lifting a load vertically. The purpose of this lesson is to establish a baseline design, observe and document system performance, and identify l;imitations related to torque, speed, friction, stability, and reliability. Through guided inquiry, students will begin posing testable questions about how modifying specific components may improve performance later trials.

(12C modifications: emphasis on observing performance differences and practical effectiveness of design choices, with less emphasis on formal theoretical justification)

STEP 1: CURRICULUM CONNECTIONS

Ontario Curriculum Overall Expectations (numbers from documents and details)	**Ontario Curriculum Specific Expectations** Numbers from documents and details -- a realistic number of expectations (1 or 2), connect to assessment.
Overall Curriculum expectations are all referenced in the main document	Specific Curriculum expectations are all referenced in the main document
Learning Goals **Discuss with learners:** ***What will I be learning today?*** Clearly identify what learners are expected to know and be able to do.	**Success Criteria** **Discuss with learners:** ***How will I demonstrate what I have learned?*** Teacher/learner identify 'look fors' in language that learners can readily understand.
We are learning to … - Design and test a mechanical elevator system using gears and rotational components - Analyze how design choices influence force, speed, stability, and mechanical advantage - Collect and use observational evidence to identify design limitations and guide improvements	I can … - Construct a working gear driven elevator system that lifts a load vertically - Describe how effectively my elevator lifts a laid in terms of speed, ease of lifting, and stability - Identify at least one limitation in my design and explain why it may be occurring - Record clear observations and measurements that can be used justify future design changes *(success criteria may be demonstrated orally, visually, through teacher conferencing))*

STEP 2: ASSESSMENT

Indicate purpose of the assessment : **[X] *FOR*** **[X] *AS*** **[ ] *OF***

Indicate Achievement Chart categories being assessed
[X] *Knowledge and Understanding* **[X] *Thinking*** **[X] *Application*** **[X] *Communication***

Indicate Learning Skills/Work Habits:
[X] responsibility, [X] organization, [ ] independent work, [X] collaboration, [X] initiative, [X] self-regulation
(Incorporate one with Learning Goal and Success Criteria)

Indicate Assessment Mode:	**Indicate Assessment Strategy :** What will learners do to demonstrate their learning?	**Indicate Assessment Tool:** Instrument used to record results/ document learning.
Written, Oral, Performance (Write, Say, Do) Written: Investigation hand outs will be briefly looked at for AFL. Brief feedback may be provided to students in the following lesson to help steer them on the right track. Qualitatively, students will be assessed throughout the activity on the learning skills / work habits through the use of a tracking note sheet.	- Collaboratively designing and constructing an initial elevator prototype - Testing the system under consistent conditions - Responding to teacher questioning during investigation - Completing the elevator investigation handout (Trial 1) ith observations and preliminary analysis	- Teacher anecdotal notes - Investigation handout

STEP 3: CONSIDERATIONS FOR PLANNING

Prior Learning: What prior experiences, knowledge and skills do the learners bring with them to this learning experience?
Students bring prior understanding of:
- forces and motion
- Simple machines (levels, pulleys, gears)
- Basic concepts of work and energy

(Some students may also have experiential knowledge from technology or construction based contexts)

Differentiation -- How will I differentiate the instruction to ensure the inclusion of all learners? (Choose 1 or 2 areas in Year 1)

Content
- Key concepts are revisited through physical manipulation and observation

Process
- Structured checkpoints, guiding questions, and teacher conferencing support gradual release

Product
- Students may demonstrate understanding through diagrams, written observations, or oral explanations

Environment
- collaborative, hands-on workplace supports peer learning and risk taking

Resources and Materials & Technology Integration -- List ALL items necessary for the delivery of the lesson. Include any attachments of learner worksheets used and teacher support material that will support communication of instruction.
Per group:
- Cardboard or foam board
- Gears of varying sizes
- Skewers or dowels

- String or yarn
- Tape, scissors, ruler
- Masses (washers, coins, weights)
- Stopwatch or phone timer
- Investigation handout package

Three Part Lesson

Indicate Instructional Strategy -- modelled, shared, guided, independent instruction, small group, conferencing, whole class, other _____

Small group: Students will be collaborating in small groups together to discuss ideas apply ideas through a design challenge
Whole class: instructions will be communicated to the whole class prior to the start of the activity

Minds on: Motivational Hook/engagement /Introduction (approximately 10-20%)
How will I engage the learners (motivational strategy, hook, activation of learners prior knowledge, activities, procedures, compelling problem)?

Teacher shows a simple elevator example (real or video) and asks:

- What parts do you think make this elevator easier or harder to lift?
- If an elevator struggled to lift a load, what components might an engineer redesign

Students redesign briefly discuss ideas, activating prior knowledge of gears, force, and motion

The video below is the one we would watch as a class to gather an understanding for how elevators systems work and some of their respective parts:
How do Elevators work (Animation)
https://www.youtube.com/watch?v=CvY-G2FTbGM

This video will either be given to students to watch in their groups after they have been selected or if there is time shown after the previous video with all questions/misconceptions have been answered. This video highlights some of the key forces at work in an elevator system which will aid students in completing their Free Body Diagrams:
Would you weigh less in an elevator? - Carol Hedden
https://www.youtube.com/watch?v=2k-2IlJcaHg

Prompting Questions and Possible Learner Responses:
Questions used to activate prior knowledge and develop critical/creative thinking

What parts do you think make this elevator easier or harder to lift?

1. If an elevator struggled to lift a load, what components might an engineer redesign first?
2. When you turn a handle to lift a load, what is the input and what is the output?
3. If we make the system easier to lift, what trade-off might we see?
4. What does “mechanical advantage” mean in your own words?
5. Two systems lift the same load. One needs a lot of effort but lifts quickly. The other needs little effort but lifts slowly. Which one has higher mechanical advantage? Explain
6. What factors besides gear sizes might cause the elevator to “fail” even if the ratios should help?

Action: During /Working on it (approximately 60%)
How will I provide practice of new concepts, and have them demonstrate new learning?

Students work in small groups to design and construct an inguinal gear driven elevator system. Groups are explicitly told that any component may be adjusted, including:

- Gear size configuration
- Spool diameter
- Axle placement and alignment
- Cable path attachment
- Structural supports

Prompting Questions and Possible Responses:
Critical thinking questions used throughout the lesson

1. What are your group's initial thoughts towards this activity? Do you have a pre conceived idea of what you already want to change?
2. What steps is your group taking to ensure that work is distributed fairly?

Students then conduct a baseline trial, testing whether the elevator can lift the load and how effectively it does so. Students record observations in their handout package, including: - Ease of lifting (qualitative force required) - Speed of movement - smoothness and stability - Evidence of friction	3. How is your group ensuring that work and data being collected is done fairly and truthfully? 4. What is the importance of drawing a Free Body Diagram for this activity? Why might it be important in a real world application? 5. Why did your group decide to design your initial lift like you did? 6. What does your group think the hardest part of this activity is going to be? 7. How does your Free body Diagram explain the motion observed in your system?
Consolidation & Connection (Reflect and Connect) (approximately 10 - 20%) How will I bring all the important ideas from the learning experiences together for/with learners? How will I check for understanding?	
Groups reflect and identify: - One specific weakness in their current design - One proposed modification, supported by evidence from observations Students share ideas during a brief whole-class discussion allowing groups to hear multiple approaches.	**Prompting Questions:** Questions used to assess learners' understanding State one specific weakness in your current design (not vague). What exactly happened? 1. What evidence from your trial supports that weakness? 2. Choose one modification you will try for the next class. Describe the change in one or two sentences. 3. Link your modification to a cause: what do you think is causing the weakness? 4. If your weakness is friction, what modification targets friction directly (not gear ratio)? 5. If your weakness is instability, what modification targets stability directly? 6. Compare (quick share-out): which groups prioritized speed, which prioritized ease, and which prioritized reliability? 7. What will you keep the same in lesson 2 so you can make a valid comparison to today's baseline. 8. Exit Question: In a sentence or two, explain how a design choice influenced the performance today
Extension Activities -- What will learners do when work is completed? What will learners do if they finish early?	
Groups that complete testing early may: - Begin implementing their proposed modification - Sketch alternative gear or spool configurations - Predict how their modification might change performance	
Next Steps -- Where will this lesson lead to next?	
In the next lesson, students redesign and test one or more components of their elevator system, comparing results to their baseline trial and continuing the inquiry cycle.	

LESSON PLAN

UPDATED 12 June 2019

Name: **Cohort:**

Lesson Plan Day 2

Lesson Title: How Do Gear Ratios Influence, Torque, Speed, and Mechanical Advantage?	**Date:**
Grade Level: 11U *(12C Modifications)*	**Subject/Strand: Physics**
Topic: Gears, Torque, Mechanical Advantage	**Length of Period: 75 minutes**

Lesson Plan Description -- What do I want the learners to know and/or be able to do? What are the big ideas/enduring understandings for the lesson/unit? Indicate using 1-2 sentences.

Students will intentionally redesign and modify one or more components of their gear driven elevator system based on evidence gathered during their baseline trial. Students will conduct a controlled comparison between thri original and modified designs collecting quantitative and qualitative data to evaluate whether performance has improved. Through guided inquiry, students will use mathematical evidence to justify whether their design modification successfully addressed an identified limitation.

(Emphasis is placed on practical effectiveness, observable improvement, and real-world relevance of the modification, with scaffolded or qualitative treatment of calculations)

STEP 1: CURRICULUM CONNECTIONS

Ontario Curriculum Overall Expectations (numbers from documents and details)	**Ontario Curriculum Specific Expectations** Numbers from documents and details -- a realistic number of expectations (1 or 2), connect to assessment.
Overall Curriculum expectations are all referenced in the main document	Specific Curriculum expectations are all referenced in the main document
Learning Goals **Discuss with learners: *What will I be learning today?*** Clearly identify what learners are expected to know and be able to do.	**Success Criteria** **Discuss with learners: *How will I demonstrate what I have learned?*** Teacher/learner identify 'look fors' in language that learners can readily understand.
We are learning to … - Improve the performance of a mechanical system by making evidence based design modifications - Apply principles of forces and energy to explain changes in system behaviour - Compare experimental trials using fair testing principles and mathematical reasoning	I can … - Clearly describe what i changed in my elevator system and why - Test my modified design using the same load and procedure as my baseline trial - Compare performance data (speed, force, stability, friction, efficiency) between designs - Justify whether my modification improved the system using evidence and physics concepts *(success criteria may be demonstrated through oral explanation, annotated diagrams, or teacher conferencing rather than formal calculations)*

STEP 2: ASSESSMENT

Indicate purpose of the assessment : **[X] *FOR*** **[X] *AS*** **[ ] *OF***

Indicate Achievement Chart categories being assessed
[X] *Knowledge and Understanding* **[X] *Thinking*** **[X] *Application*** **[X] *Communication***

Indicate Learning Skills/Work Habits:
[X] responsibility, [X] organization, [ ] independent work, [X] collaboration, [X] initiative, [X] self-regulation
(Incorporate one with Learning Goal and Success Criteria)

Indicate Assessment Mode:	**Indicate Assessment Strategy :** What will learners do to demonstrate their learning?	**Indicate Assessment Tool:** Instrument used to record results/ document learning.
Written, Oral, Performance (Write, Say, Do)	- Modifying one or more components - Conducting a fair comparison trial using the same load and testing conditions - Completing the comparison section of the investigation handout - Responding to teacher questions during testing and analysis	- Teacher anecdotal notes - Investigation handout (Trials 2 and comparison analysis)

STEP 3: CONSIDERATIONS FOR PLANNING

Prior Learning: What prior experiences, knowledge and skills do the learners bring with them to this learning experience?
Students should be bring:
- Observations and data from the baseline trial
- Prior understanding of forces, torque, motion, and simple machines
- Initial hypotheses about how design changes may influence performance

Differentiation -- How will I differentiate the instruction to ensure the inclusion of all learners? (Choose 1 or 2 areas in Year 1)

Content
- Mathematical relationships are emphasized for Grade 11U, practical interpretation is emphasized for Grade 12C

Process
- Structured checkpoints ensure all groups articulate their reasoning before testing

Product
- Students may communicate understanding through calculation, graphs, diagrams, or oral explanations

Environment
- collaborative , kinesthetic setting supports peer learning and iterative designs

Resources and Materials & Technology Integration -- List ALL items necessary for the delivery of the lesson. Include any attachments of learner worksheets used and teacher support material that will support communication of instruction.
Per group:
- Previously constructed elevator system
- Any modification components they may need (gears of varying sizes, additional pulleys, spools, et.)
- Stopwatch or phone timer
- Masses
- Investigation handout package

Three Part Lesson

Indicate Instructional Strategy -- modelled, shared, guided, small group, conferencing, whole class

Minds on: Motivational Hook/engagement /Introduction (approximately 10-20%)
How will I engage the learners (motivational strategy, hook, activation of learners prior knowledge, activities, procedures, compelling problem)?

Students regroup and review their baseline trial results from Day 1. Groups will briefly discuss their planned modification before beginning construction

Prompting Questions and Possible Learner Responses:
Questions used to activate prior knowledge and develop critical/creative thinking

1. Has the break given your group any new ideas?
2. Now that you are looking at your data with fresh eyes, are you noticing anything that you might have missed yesterday?
3. Moving forward, what factor(s) might limit or restrict your design ideas?
4. How can your analytical evidence be used to help determine what part of your lift systems to change?
5. Why is initial data so important when conducting real world experiments?

Action: During /Working on it (approximately 60%)
How will I provide practice of new concepts, and have them demonstrate new learning?

Students:
- Implement on intentional design modification
- Re-test their elevator using the same load and procedure
- Record performance data and observations
- Compare Trial 1 and Trial 2 results using tables, graphs, and calculations

Grade 11U:
- Calculating average speed
- Estimating work input/poutout
- Commenting on efficiency and sources of energy loss

Grade 12C:
- Qualitative comparison of ease lifting, smoothness, and practicality
- Emphasis on real-world design trade offs

Prompting Questions and Possible Responses:
Critical thinking questions used throughout the lesson

1. What was your biggest limitation from Trial 1, stated as a specific observable problem?
2. What ONE modification are you testing today, and what exact part(s) are changing (gear sizes, spool diameter, axle alignment, cable path, supports)?
3. How does your modification directly target yesterday's limitation (cause -> effect)?
4. What do you predict will change in performance (time, turns, ease of lifting, stability, friction evidence), and why?
5. What variables must stay identical to make this a fair comparison to your baseline trial?
6. What will count as "improved performance" for your group today (choose 1-2 metrics you will prioritize)?
7. What data will you collect during your trials so your comparison is valid and easy to justify?
8. What trade-offs are you expecting (easier but slower, faster but harder, more table but heavier, etc.)?
9. Who is responsible for: builder adjustments, timer/turn counter, recorder, and test operator (to reduce inconsistency)?

Consolidation & Connection (Reflect and Connect) (approximately 10 - 20%)

How will I bring all the important ideas from the learning experiences together for/with learners? How will I check for understanding?

Groups reflect and decide whether to:

- implement a second modification (if one has not already been done, or
- Begin preparing to communicate and justify their final design

Students complete a brief reflection sections in their handout summarizing:

- What improved
- What did not
- What evidence support their conclusion

Prompting Questions:

Questions used to assess learners' understanding

1. Did your modification improve the system based on your chosen success criteria? State yes/no and cite your best data point
2. What is your strongest piece of evidence that the change worked (or did not work)?
3. What trade-off did you create (what improved, what got worse, and what stayed the same)?
4. What physics idea best explains your results today (torque, mechanical advantage, work/energy losses, friction)?
5. If your data is messy, what is one change to your testing method that would improve reliability next class?
6. Decide: do you make one additional change next, or do you shift to preparing to communicate your findings? Why?
7. If you will make another change, what will you change next and what do you predict will happen?
8. If you will communicate findings, what 2 visuals or data displays will best support your claim (table, before/after comparison, simple graph)?
9. Exit check: in one sentence, what did you change, and in one sentence, what did the evidence show?

Extension Activities -- What will learners do when work is completed? What will learners do if they finish early?

Groups who complete testing early may:

- Begin drafting their presentation or design pitch
- Sketch alternative designs they did not have time to test
- Connect their findings to real-world elevators, lifts, or mechanical systems

Next Steps -- Where will this lesson lead to next?

Students will either continue modifying their elevator design or shift focus towards finalizing and communicating their design decisions through a presentation (pitch) that explains how physics principles informed their solutions.

LESSON PLAN

UPDATED 12 June 2019

Name: **Cohort:**

Lesson Plan Day 3

Lesson Title: How Do Gear Ratios Influence, Torque, Speed, and Mechanical Advantage?	**Date:**
Grade Level: 11U *(12C Modifications)*	**Subject/Strand: Physics**
Topic: Gears, Torque, Mechanical Advantage	**Length of Period: 75 minutes**

Lesson Plan Description -- What do I want the learners to know and/or be able to do? What are the big ideas/enduring understandings for the lesson/unit? Indicate using 1-2 sentences.

Students will finalize testing of their elevator system and/or consolidate evidence from previous trials, then shift their focus toward explaining and justifying how and why their final design performs as it does. Emphasis is placed on constructing clear, logical explanations that connect observations, data, and physics concepts, and on preparing to communicate these ideas effectively to an audience. This lesson marks the transition from experimentation to knowledge construction and communication.

(explanations may be more descriptive and demonstration based, emphasizing observable outcomes, design practicality, and real-world relevance rather than formal calculations)

STEP 1: CURRICULUM CONNECTIONS

Ontario Curriculum Overall Expectations (numbers from documents and details)	**Ontario Curriculum Specific Expectations** Numbers from documents and details -- a realistic number of expectations (1 or 2), connect to assessment.
Overall Curriculum expectations are all referenced in the main document	Specific Curriculum expectations are all referenced in the main document
Learning Goals **Discuss with learners: *What will I be learning today?*** Clearly identify what learners are expected to know and be able to do.	**Success Criteria** **Discuss with learners: *How will I demonstrate what I have learned?*** Teacher/learner identify 'look fors' in language that learners can readily understand.
We are learning to … - Explain how specific design choices affect the performance of a mechanical system - Organize evidence and physics concepts into a coherent explanation - Communicate scientific reasoning clearly and collaboratively	I can … - Justify my final elevator design using data, observations, and physics principles - Explain how different components influence performance - Use diagrams, graphs, or calculations to support my explanation - Work collaboratively to prepare a clear, well organized presentation *(success criteria may be met through explanation, physical demonstration, annotated visuals, or teacher conferencing)*

STEP 2: ASSESSMENT

Indicate purpose of the assessment : **[ ] *FOR*** **[X] *AS*** **[ ] *OF***

Indicate Achievement Chart categories being assessed
[X] *Knowledge and Understanding* **[X] *Thinking*** **[X] *Application*** **[X] *Communication***

Indicate Learning Skills/Work Habits:
[X] responsibility, [X] organization, [ ] independent work, [X] collaboration, [X] initiative, [X] self-regulation
(Incorporate one with Learning Goal and Success Criteria)

Indicate Assessment Mode: *Written, Oral, Performance* (Write, Say, Do)	**Indicate Assessment Strategy :** What will learners do to demonstrate their learning? - Organizing experimental results into claims, evidence, and reasoning - Creating visual supports (diagrams, graphs, tables) - Participating in teacher conferencing to clarify and refine expectations - Providing and receiving peer feedback using a structured reflection handout	**Indicate Assessment Tool:** Instrument used to record results/ document learning. - Teacher anecdotal notes - Investigation handout (analysis and reflection sections) - Peer feedback handout - Draft presentation materials

STEP 3: CONSIDERATIONS FOR PLANNING

Prior Learning: What prior experiences, knowledge and skills do the learners bring with them to this learning experience?
Students will draw on:
- Data and observation from baseline and modified trials
- Prior knowledge of forces, torque, motions, and energy
- Experience with inquiry, fair testing, and evidence based reasoning

Differentiation -- How will I differentiate the instruction to ensure the inclusion of all learners? (Choose 1 or 2 areas in Year 1)

Content
- Formal calculation and physic terminology emphasized for Grade 11U, conceptual explanations emphasized for Grade 12C

Process
- Structured checkpoints guided students from raw data to explanations

Product
- Understanding may be demonstrated through graphs, diagrams, oral explanations, or demonstrations

Environment
- Collaborative workspace supports discussion, questioning and revision

Resources and Materials & Technology Integration -- List ALL items necessary for the delivery of the lesson. Include any attachments of learner worksheets used and teacher support material that will support communication of instruction.
- Completed elevator systems
- Investigation handout packages
- Chart paper or digital slides
- rulers/markers
- Peer feedback/reflection handouts

Three Part Lesson

Indicate Instructional Strategy -- modelled, shared, guided, small group, conferencing, whole class

Minds on: Motivational Hook/engagement /Introduction (approximately 10-20%)
How will I engage the learners (motivational strategy, hook, activation of learners prior knowledge, activities, procedures,

compelling problem)?

Students regroup and are resented tight the framing challenge: - Imagine your elevator design is being considered for use in a real community building. You must persuade decision makers that your design is effective, safe, and worth implementing Groups briefly discuss how they want to frame their explanation or pitch.	**Prompting Questions and Possible Learner Responses:** Questions used to activate prior knowledge and develop critical/creative thinking 1. What is a one sentence claim about your final design's performance? 2. What is you audience ofr the pitch, and what do they care about most(speed, ease, reliability, safety)? 3. Which one or two data metrics best support your claim (time, turns, ease rating, success rate)? 4. What is the most important trade-off your design shows (what improved, what got worse)? 5. What design feature caused that trade-off (gear ratio choice, spool size, alignment, supports)? 6. What is the simplest diagram you can make so someone can understand your system quickly 7. What is the one physics idea you will use to explain your result? 8. What will you show as "before vs after" so your improvement is obvious 9. Who is responsible today for: graph/table, diagram, CER writing, speaking plan?
Action: During /Working on it (approximately 60%) How will I provide practice of new concepts, and have them demonstrate new learning?	
Students work to: - Finalize any remaining testing (if needed) - Organize data into clear claims and supporting evidence - Create diagrams, graphs, and/or calculations that illustrate system behaviour - Begin assembling their representation or design pitch Grade 11U: - Explicit connections between evidence and physics concepts - Use of calculations, graphs, and correct scientific terminology Grade 12C: - Emphasis on practical performance, safety, efficiency, and usability - Explanations supported by demonstrations and annotated visuals	**Prompting Questions and Possible Responses:** Critical thinking questions used throughout the lesson 1. If you were to run one final controlled test, what variable would you isolate, and why is isolating that variable important for validity? (Responses would vary) 2. How have you ensured that your results are reliable and support your evidence? 3. What single data point do you think best supports the performance of your lift? 4. How could someone from another group challenge the validity of your findings? 5. What might the slope of your graph represent? Could it represent more than one relationship? 6. How do your efficiency calculations reflect energy losses in your elevator? 7. Did your group find that there were certain aspects that numbers could not account for

	in making your lift system? 8. If we were to remove friction entirely, how would that affect your results? Would it change your redesign?
Consolidation & Connection (Reflect and Connect) (approximately 10 - 20%) How will I bring all the important ideas from the learning experiences together for/with learners? How will I check for understanding?	
Students complete peer feedback and reflection handouts, focusing on: - Clarity of explanations - Strength of evidence - Effectiveness of communication Groups discuss feedback and identify one specific improvement to make before final presentations	**Prompting Questions:** Questions used to assess learners' understanding 1. How does your design reflect real world engineering priorities? 2. What piece of data does your group think would best persuade a decision maker? 3. What is the most effective way to split up the "pitch" so that its participation is fairly distributed? 4. What did your group do to ensure that everyone had a role and participated fairly? 5. If you had unlimited time and resources, what would be your group's next improvement step that would push this lift system over the top?
Extension Activities -- What will learners do when work is completed? What will learners do if they finish early?	
Groups who finish early amy: - Rehearse their presentation, assign speaking roles and timing - Refine visuals or explanations for clarity - Anticipate questions and prepare responses	
Next Steps -- Where will this lesson lead to next?	
Students will finalize their presentations and deliver a design pitch to the class, clearly explaining how physics principles informed their final elevator design and why it represents an effective solution	

LESSON PLAN

UPDATED 12 June 2019

Name: **Cohort:**

Lesson Plan Day 4

Lesson Title: How Do Gear Ratios Influence, Torque, Speed, and Mechanical Advantage?	**Date:**
Grade Level: 11U *(12C Modifications)*	**Subject/Strand: Physics**
Topic: Gears, Torque, Mechanical Advantage	**Length of Period: 75 minutes**

Lesson Plan Description -- What do I want the learners to know and/or be able to do? What are the big ideas/enduring understandings for the lesson/unit? Indicate using 1-2 sentences.

Students will present and define their final elevator system designs, explaining how gear ratios, torque, and mechanical advantage influence system performance. They will communicate scientific reasoning using evidence from testing and reflect on both their conceptual learning and collaborative process. This lesson culminates the inquiry cycle by emphasizing knowledge mobilization, reflection and evaluation reinforcing that learning in science extends beyond correct answer to include justification, communication, and teamwork.

(presentation may emphasize physical demonstration, real-world application, safety, and design, practically, with reduced emphasis on formal calculations)

STEP 1: CURRICULUM CONNECTIONS

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Learning Goals **Discuss with learners: *What will I be learning today?*** Clearly identify what learners are expected to know and be able to do.	**Success Criteria** **Discuss with learners: *How will I demonstrate what I have learned?*** Teacher/learner identify 'look fors' in language that learners can readily understand.
We are learning to … - Communicate scientific thinking clearly and accurately - Justify design decisions using evidence and physics principles - Reflect on out learning, problem solving, strategies, and teamwork	I can … - Clearly explain how my elevator system works - Use data, observation, and physical concepts to justify design choices - Respond thoughtfully to questions about my design - Reflect honestly and specifically on my learning and group contributions *(Success may be demonstrated through oral explanation, demonstration, annotated diagrams, or conferencing rater than formal written justification)*

STEP 2: ASSESSMENT

Indicate purpose of the assessment : **[ ] *FOR*** **[X] *AS*** **[X] *OF***

Indicate Achievement Chart categories being assessed
[X] *Knowledge and Understanding* [X] *Thinking* [X] *Application* [X] *Communication*

Indicate Learning Skills/Work Habits:
[X] responsibility, [X] organization, [ ] independent work, [X] collaboration, [X] initiative, [X] self-regulation
(Incorporate one with Learning Goal and Success Criteria)

Indicate Assessment Mode:	**Indicate Assessment Strategy :** What will learners do to demonstrate their learning?	**Indicate Assessment Tool:** Instrument used to record results/ document learning.
Written, Oral, Performance (Write, Say, Do)	- Delivering a structures group presentation or pitch - Responding to questions that probe scientific reasoning - Completing and individual self-reflection - Completing a peer/group evaluation focused on the RAT process	- Teacher anecdotal notes - Presentation rubric - Individual self-reflection handout - peer/group evaluation handout

STEP 3: CONSIDERATIONS FOR PLANNING

Prior Learning: What prior experiences, knowledge and skills do the learners bring with them to this learning experience?
Student bring:
- Experimental data from multiple elevator trials
- Prior understanding of forces, torque, gears, and mechanical advantage
- Experience with inquiry based investigation and collaborative problem solving

(some students may also have technological or applied experience with mechanical systems)

Differentiation -- How will I differentiate the instruction to ensure the inclusion of all learners? (Choose 1 or 2 areas in Year 1)

Content
- Theoretical explanations emphasized for Grade 11U, practical and conceptual explanations emphasized for Grade 12C

Process
- Guiding questions and checkpoints scaffold explanations

Product
- Learning demonstrated through presentations, diagrams, reflections, or demonstrations

Environment
- Collaboration, respectful learning space that values multiple forms of expression

Resources and Materials & Technology Integration -- List ALL items necessary for the delivery of the lesson. Include any attachments of learner worksheets used and teacher support material that will support communication of instruction.\
- Completed elevator systems
- Presentation (slides, chart paper, pamphlets, etc.)
- Reflection and evaluation handouts
- Rubric (student friendly version shared in advance)

Three Part Lesson

Indicate Instructional Strategy -- modelled, shared, guided, small group, conferencing, whole class

Minds on: Motivational Hook/engagement /Introduction (approximately 10-20%)
How will I engage the learners (motivational strategy, hook, activation of learners prior knowledge, activities, procedures, compelling problem)?

Groups are given 10-15 minutes to: - Rehearse their presentation - Decide who will speak to each aspect of the design - Ensure that evidence and reasoning are clearly connected	**Prompting Questions and Possible Learner Responses:** Questions used to activate prior knowledge and develop critical/creative thinking ***Probably don't really need prompting questions here / could be more of a checklist for students but don't want to psych them out prior to presenting*** 1. What is your hook for the first 10 seconds (what problem are you solving)? 2. What is your one-sentence claim (say it exactly as you will present it)? 3. What is the order of your pitch (system overview → change → data → explanation → trade-off)? 4. What visual will you point to when you state your claim (diagram, table, graph)? 5. What are your two key numbers you must say out loud during the pitch? 6. What is your single trade-off statement (improved ___ but reduced ___)? 7. Who is presenting which section, and what is each person's exact first sentence? 8. What is your time target per speaker so you finish cleanly? 9. What question are you most likely to get, and what evidence will you use to answer it?
Action: During /Working on it (approximately 60%) How will I provide practice of new concepts, and have them demonstrate new learning?	
Students pitch their modified elevator system to the class. Presentations include: - A brief overview of the design - Explanation of key modifications (gear ratio, spool size, pulleys, etc.) - Evidence from testing (data, observations, graphs) - Explicit connections to torque, force, speed, and mechanical advantage *Assuming 30 students in the class, can do groups of 5-6 where each group will have 6-8 minutes to present. 3 minutes will be given to each group for questions alongside setting up for the next presentation. This should be the majority of the class.*	**Prompting Questions and Possible Responses:** Critical thinking questions used throughout the lesson ***Prompting questions should align with the rubric / the expectations for which student are being marked on.*** 1. What does your elevator system do, in one clear sentence? 2. What was your original limitation from early trials (specific observable issue)? 3. What exactly did you change (component + direction of change)? 4. What stayed the same during testing so your comparison is fair? 5. What was your baseline value for your key metric? 6. What was your modified value for the same metric? 7. What is the difference and percent change? 8. What observation supports the numbers (stability, friction, slipping, stalling)? 9. What physics idea explains why your change affected performance? 10. What trade-off did your change create, and

	why is it acceptable (or not)? 11. If someone copied your design, what is one build detail they must get right? 12. What is one limitation of your data or method (and how would you fix it)? 13. If you had one more class, what would you test next (one variable only)? 14. What is your main takeaway about gear ratio/spool size/stability from this project? 15. Teacher question check: "How do you know the improvement wasn't just chance?" What is your evidence-based answer?
Consolidation & Connection (Reflect and Connect) (approximately 10 - 20%) How will I bring all the important ideas from the learning experiences together for/with learners? How will I check for understanding?	
Students complete: - Individual self-reflection - peer/group evaluation	**Prompting Questions:** Questions used to assess learners' understanding 1. What did your group do well in the pitch (one specific strength)? 2. What is one change you would make to improve clarity or evidence next time? 3. What did you personally contribute that mattered to the final product? 4. What is one skill you improved (testing, data recording, explaining, collaborating)? 5. What is one teamwork challenge you had, and how did you handle it? 6. What is one piece of feedback you received that you agree with, and why? 7. What is one physics concept you understand better now because of the build/testing? 8. What is one real-world situation where this type of design trade-off matters? 9. Exit reflection: In one sentence, what did the evidence show about your final design?
Extension Activities -- What will learners do when work is completed? What will learners do if they finish early?	
If presentations conclude early: - Students complete reflections in greater depth - Groups may revise explanations based on peer feedback - Students may annotate diagrams or visuals to further clarify thinking - Student may share peer/group feedback with group - Work on other work from other classes	
Next Steps -- Where will this lesson lead to next?	
This serves as a stepping stone into the broader Mechanical Systems unit. Building on their elevator investigation, s students will extend their understanding of how individual components interact within complex systems. *(Students will focus on identifying mechanical systems in real-world contexts, such as elevators, cranes, bicycles, and industrial machinery, and evaluating how design choices affect usability, safety, and efficiency rather than formal*	

calculations)